

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)			Reference to the ratio of output power to input power or the efficiency equation (1) Output power becomes a smaller fraction of the input power (1) so agree. Alternative 1: 100 % efficient up to about 46 V (1) Then output power is less than input power (1) so agree. Alternative 2: Minimum of 2 efficiency calculations (2) so agree. To award both marks the conclusion must be present.			2	2		2
	(b)			Substitution: $I = \frac{P}{V} = \frac{180}{100}$ (1) = 1.8 [A] (1)	1	1		2	2	2
	(c)			Less heat loss (1) accept eddy currents reduced or no heat loss or prevents overheating More efficient (1)	2			2		
	(d)			[High voltage] results in low current (1) Don't accept low amps <u>Less</u> heating effects in wires or <u>more</u> efficient transmission (1) Don't accept prevents or no heat loss	2			2		
				Question 8 total	5	1	2	8	2	4